

Asfi Hassan

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EDUCATION

University of Windsor

Bachelor of Computer Science with Honour's

Windsor, ON

Sep. 2023 – Present

WORK EXPERIENCE

Researcher (Software Development/Number Theory) | *University of Windsor*

May 2026 – Aug. 2026

- Selected for the NSERC Undergraduate Student Research Award (USRA) to conduct computational number theory research under Dr. Ahmed Biniiaz.
- Developed an interactive Python application using Tkinter and Matplotlib to visualize runner trajectories.
- Implemented algorithms to generate, analyze, and visualize runner configurations from dynamic user inputs.
- Designed modular software architecture to support experimentation, visualization, and algorithm evaluation.
- Applied computational methods to investigate the Lonely Runner Conjecture.

Program Administration Support | *Ministry of the Solicitor General*

Sep. 2025 – Dec. 2025

- Oversaw the flow of information, tracked documents, and maintained accurate records using internal online applications in a highly secure and confidential government building.
- Gathered and verified information, liaised with appropriate contacts to obtain clarification or share data across departments.
- Established and maintained a structured filing system for confidential materials and operational documents.
- Prepared various correspondence, including letters, memos, reports, charts, graphs, and standing orders, ensuring accuracy and consistency.
- Deployed software and operating system upgrades across institutional devices, ensuring compatibility, application stability, and secure VPN protocol configuration while providing basic first-level IT support to coworkers.
- Utilized a confidential database to query and retrieve specific information, supporting administrative and operational decision-making.

Student Office Assistant | *Ministry of the Solicitor General*

May 2025 – Aug. 2025

- Prepared and organized correspondence, performed data entry and word processing functions, and created a variety of documents.
- Assisted with archival and filing projects by following rules for file retention and destruction, using technology to manage documents.
- Conducted research, performed analysis, and input data into internal systems.
- Deployed Software and OS upgrades across organizational systems, ensuring compatibility, application stability, and secure VPN protocol integration.
- Effectively managed files and priorities by organizing records, verifying data accuracy, and resolving issues under tight deadlines.

PROJECTS

AI Expense Tracker | *Java, HTML, CSS, JavaScript, VS Code*

Dec. 2025

- Built an AI-enhanced web + Java application to track income and expenses with smart category prediction.
- Implemented automatic balance calculation, category summaries, spending analytics, and insights.
- Designed an intuitive UI and modular architecture for scalability and maintainability.

File Mapping Software Engineering Project | *Python, SCRUM/Agile, UI Prototype*

Dec. 2025

- Developed a Python-based system capable of mapping version changes across multiple programming languages such as Java and C.
- Produced a full formal Software Engineering report covering design, methodology, evaluation, and accuracy testing on a real dataset.
- Designed a UI prototype demonstrating workflow visualization and user interaction for future implementation.

EXTRA-CURRICULAR ACTIVITIES

Technician / Programmer <i>FIRST Robotics Team</i>	Nov. 2022 – April 2023
- Enhanced the robot’s autonomous functionality, earning judges’ recognition in competitions. - Programmed and optimized the robot for competition readiness. - Collaborated with peers to improve robot precision for local and regional contests.	

CERTIFICATIONS

Certified Python Developer <i>W3Schools</i>	Dec. 2024
Certified Java Developer <i>W3Schools</i>	Dec. 2024
Certificate of Continuing Education Completion <i>Cybrary</i>	July 2024

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, HTML/CSS, JavaScript
Tools: GitHub, Docker, VS Code, Vim, PyCharm, Eclipse, MS Office, Excel, Word
Operating Systems: Windows, Unix/Linux (Ubuntu, Cygwin)